ECE 647: Lecture 15

Last lecture Clarifications

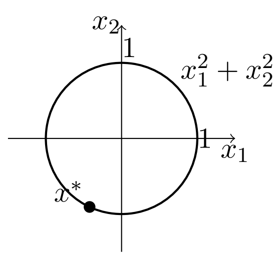
Example 1

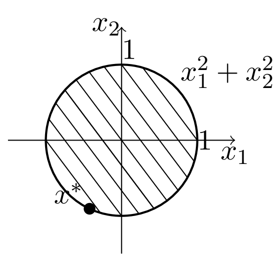
Equality constraints can sometimes be converted into inequality constraints, and vice versa, if we know additional structure of problem.

$$\begin{array}{cl} \min & x_1 + x_2 \\ \text{s.t.} & x_1^2 + x_2^2 = 1 \end{array}$$

$$\Updownarrow$$

$$\begin{array}{cl} \min & x_1 + x_2 \\ \text{s.t.} & x_1^2 + x_2^2 \leq 1 \end{array}$$





Works the same for:

$$\begin{array}{cl} max & x_1 + x_2 \\ \text{s. t.} & x_1^2 + x_2^2 = 1 \end{array}$$

Example 2

$$\min \quad f_0(g(x))$$
$$\Updownarrow$$
$$\begin{aligned} \min_{x,y} \quad & f_0(y) \\ \text{s.t.} \quad & g(x) = y \end{aligned}$$

$\Updownarrow$ f_0 is increasing,
minimum of $g(x)$ is optimal.

$$\begin{aligned} \min_{x,y} \quad & f_0(y) \\ \text{s.t.} \quad & g(x) \leq y \end{aligned}$$

ps: $g(x) = y$
$g(x) < y \rightarrow$ decrease y to improve $f_0(y)$

Directional Derivative.

Define the directional derivative of a function f at $\bar{x}$ in a direction $d \in E$ as:

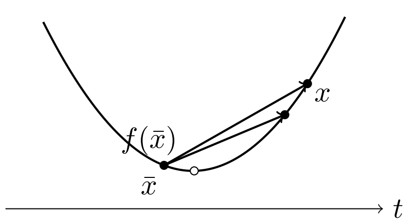
$$f'(\bar{x}; d) = \lim_{t \to 0} \frac{f(\bar{x}+td)-f(\bar{x})}{t}$$

f is differentiable at $\bar{x}$ if $f'(\bar{x}; d) = \langle a, d \rangle = a^T d$, meaning $\nabla f(\bar{x}) = a$.

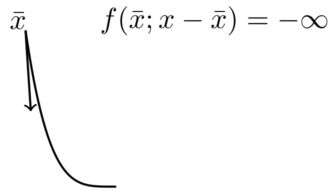
Existence of Directional Derivative for Convex Functions.

Suppose $C \subset E$ is convex, and the function $f : C \rightarrow R$ is convex. Then for any points $\bar{x}$ and x in C, the directional derivative $f'(\bar{x}; x - \bar{x})$ always exists in $[-\infty, +\infty)$.

- could be minus-infinite.



$\frac{f(\bar{x}+td)-f(\bar{x})}{t}$
is decreasing as $t \rightarrow 0$ since f is convex.



A necessary condition.

Suppose C is a convex set in E and the point $\bar{x}$ is a local minimizer of $f : C \to R$. Then for any $x \in C$, the directional derivative, if it exists, satisfies:

$$f'(\bar{x}; x - \bar{x}) \geq 0$$

Proof(by contradiction):

> If $f'(\bar{x}; x - \bar{x}) < 0$ for some $x \in C$, then the function value must be decreasing in the direction of $\bar{x} \to x$ for a small internal. Contradicts definition.

Note: For convex functions, directional derivative always exists:

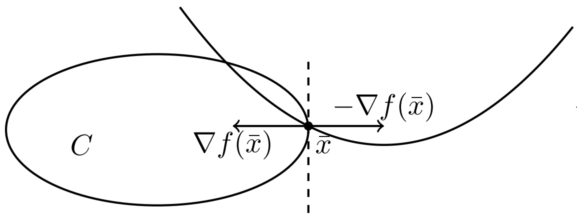
$$f'(\bar{x}; x - \bar{x}) \geq 0 \text{ for all } x \in C$$

Differentiable Functions.

If in addition f is differentiable,

$$f'(\bar{x}; x - \bar{x}) = [\nabla f(\bar{x})]^T (x - \bar{x})$$

We have:
(a) If $\bar{x} \in int(C)$, then $\nabla f(\bar{x}) = 0$.
(b) If $\bar{x}$ is at the boundary, then $-\nabla f(\bar{x})$ must point outwards.

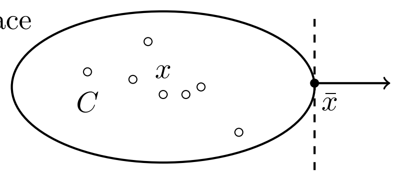


$-\nabla f(\bar{x})$ must point outwards

- Define the normal cone $N_C(\bar{x})$ to a convex set C at point $\bar{x}$ as:

$$N_C(\bar{x}) = \{d \mid \langle d, x - \bar{x}\rangle \leq 0 \quad \text{for all} \quad x \in C\}$$

smooth surface



$N_C(\bar{x})$ contains one normal vector

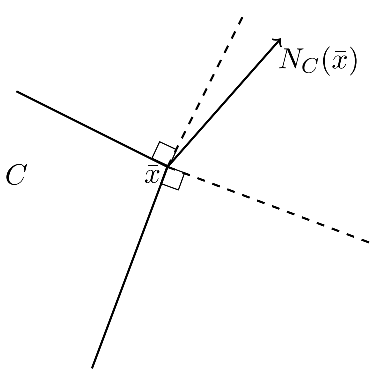
- If the constraint is defined by $f_i(x) \leq 0$:
  - At a point $\bar{x}$ such that $f_i(x) = 0$, we have:

$$[\nabla f_i(\bar{x})]^T (x - \bar{x}) \leq f_i(x) - f_i(\bar{x}) \leq 0$$

for all $x \in C$.

$\rightarrow$ $\nabla f_i(\bar{x})$ is the normal vector.

At a corner,



$N_C(\bar{x})$ is a cone that contains all normal vectors (when $\bar{x}$ is tight for multiple constraints).